

SELLER AND INVENTORY MANAGEMENT

Amazon – E-Commerce Database Management System

1.Database Design

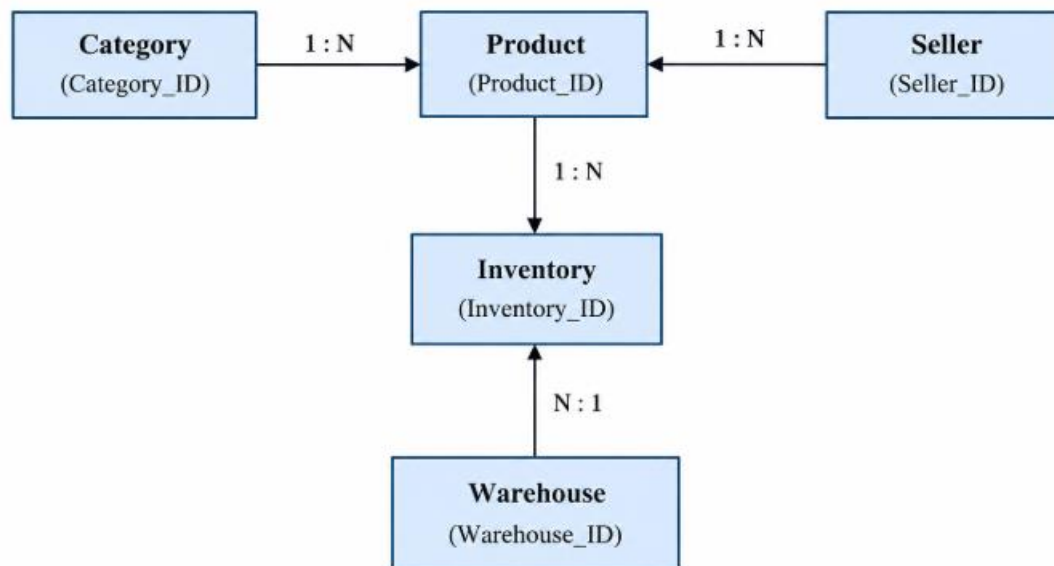
1.1 Seller Table

Attribute	Data Type	Key	Description
Seller_ID	NUMBER	PK	Unique seller ID
Seller_Name	VARCHAR2(50)	-	Seller name
Business_Name	VARCHAR2(100)	-	Business name
Email	VARCHAR2(50)	UK	Seller email
Phone_Number	VARCHAR2(10)	UK	Seller phone
GST_Number	VARCHAR2(20)	UK	GST number
Address	VARCHAR2(100)	-	Seller address
Rating	NUMBER(2,1)	-	Seller rating

1.2 Inventory Table

Attribute	Data Type	Key	Description
Inventory_ID	NUMBER	PK	Unique inventory ID
Product_ID	NUMBER	FK	References Product
Warehouse_ID	NUMBER	FK	References Warehouse
Available_Stock	NUMBER	-	Available stock quantity
Last_Updated	DATE	-	Last updated date

2. Table Relationship Diagram



3. Table Creation

3.1 creating seller table

```
CREATE TABLE Seller (  
    Seller_ID NUMBER PRIMARY KEY,  
    Seller_Name VARCHAR2(50) NOT NULL,  
    Business_Name VARCHAR2(100) NOT NULL,  
    Email VARCHAR2(50) UNIQUE,  
    Phone_Number VARCHAR2(10) UNIQUE,  
    GST_Number VARCHAR2(20) UNIQUE,  
    Address VARCHAR2(100),  
    Rating NUMBER(2,1)  
);
```

Table created.

3.2 creating Inventory Table

```
CREATE TABLE Inventory (  
    Inventory_ID NUMBER PRIMARY KEY,  
    Product_ID NUMBER,  
    Warehouse_ID NUMBER,  
    Available_Stock NUMBER,  
    Last_Updated DATE,  
    CONSTRAINT fk_inventory_product  
    FOREIGN KEY (Product_ID)  
    REFERENCES Product(Product_ID),  
    CONSTRAINT fk_inventory_warehouse  
    FOREIGN KEY (Warehouse_ID)  
    REFERENCES Warehouse(Warehouse_ID)  
);
```

Table created.

4. Seller Data Insertion

```
SQL> INSERT INTO Seller VALUES (201, 'Tech World', 'Tech World Electronics', 'techworld@gmail.com', '9876543210', 'GST201', 'Chennai', 4.5);
1 row created.

SQL> INSERT INTO Seller VALUES (202, 'Fashion Hub', 'Fashion Hub Store', 'fashionhub@gmail.com', '9876543211', 'GST202', 'Bangalore', 4.2);
1 row created.

SQL> INSERT INTO Seller VALUES (203, 'Book House', 'Book House Publishers', 'bookhouse@gmail.com', '9876543212', 'GST203', 'Chennai', 4.6);
1 row created.

SQL> INSERT INTO Seller VALUES (204, 'Home Needs', 'Home Needs Appliances', 'homeneeds@gmail.com', '9876543213', 'GST204', 'Coimbatore', 4.3);
1 row created.

SQL> INSERT INTO Seller VALUES (205, 'Beauty Care', 'Beauty Care Store', 'beautycare@gmail.com', '9876543214', 'GST205', 'Madurai', 4.4);
1 row created.

SQL> INSERT INTO Seller VALUES (206, 'Sports World', 'Sports World Store', 'sportsworld@gmail.com', '9876543215', 'GST206', 'Chennai', 4.1);
1 row created.

SQL> INSERT INTO Seller VALUES (207, 'Toy Zone', 'Toy Zone Store', 'toyzone@gmail.com', '9876543216', 'GST207', 'Salem', 4.0);
1 row created.

SQL> INSERT INTO Seller VALUES (208, 'Daily Needs', 'Daily Needs Store', 'dailyneeds@gmail.com', '9876543217', 'GST208', 'Trichy', 4.3);
1 row created.

SQL> INSERT INTO Seller VALUES (209, 'Footwear Point', 'Footwear Point Store', 'footwearpoint@gmail.com', '9876543218', 'GST209', 'Chennai', 4.2);
1 row created.

SQL> INSERT INTO Seller VALUES (210, 'Furniture World', 'Furniture World Store', 'furnitureworld@gmail.com', '9876543219', 'GST210', 'Coimbatore', 4.5);
1 row created.
```

5. Inventory Data Insertion

```
SQL> INSERT INTO Inventory VALUES (401, 101, 301, 10, SYSDATE);
1 row created.

SQL> INSERT INTO Inventory VALUES (402, 102, 302, 20, SYSDATE);
1 row created.

SQL> INSERT INTO Inventory VALUES (403, 103, 301, 30, SYSDATE);
1 row created.

SQL> INSERT INTO Inventory VALUES (404, 104, 303, 15, SYSDATE);
1 row created.

SQL> INSERT INTO Inventory VALUES (405, 105, 304, 50, SYSDATE);
1 row created.

SQL> INSERT INTO Inventory VALUES (406, 106, 305, 40, SYSDATE);
1 row created.

SQL> INSERT INTO Inventory VALUES (407, 107, 306, 25, SYSDATE);
1 row created.

SQL> INSERT INTO Inventory VALUES (408, 108, 307, 0, SYSDATE);
1 row created.

SQL> INSERT INTO Inventory VALUES (409, 109, 308, 12, SYSDATE);
1 row created.

SQL> INSERT INTO Inventory VALUES (410, 110, 309, 0, SYSDATE);
1 row created.
```

6. Seller Product Information

```
SELECT
    s.Seller_ID,
    s.Seller_Name,
    p.Product_ID,
    p.Product_Name,
    i.Available_Stock
FROM Seller s
JOIN Product p
ON s.Seller_ID = p.Seller_ID
JOIN Inventory i
ON p.Product_ID = i.Product_ID;
```

```
SELLER_ID SELLER_NAME                PRODUCT_ID
-----
PRODUCT_NAME
-----
AVAILABLE_STOCK
-----
      201 Tech World                101
Laptop
      10
      201 Tech World                102
Smartphone
      20
      201 Tech World                103
Headphones
      30
      201 Tech World                104
Smart Watch
      15
      202 Fashion Hub                105
T-Shirt
      50
      202 Fashion Hub                106
      40
      203 Book House                107
Python Programming Book
      25
      203 Book House                108
Data Science Book
      0
      204 Home Needs                109
Mixer Grinder
      12
      204 Home Needs                110
Electric Kettle
      0

10 rows selected.
```

7. Inventory Status

7.1 Display Available Products

```
SQL> SELECT
  2     p.Product_ID,
  3     p.Product_Name,
  4     i.Available_Stock
  5 FROM Product p
  6 JOIN Inventory i
  7 ON p.Product_ID = i.Product_ID
  8 WHERE i.Available_Stock > 0;
```

PRODUCT_ID	PRODUCT_NAME	AVAILABLE_STOCK
101	Laptop	10
102	Smartphone	20
103	Headphones	30
104	Smart Watch	15
105	T-Shirt	50
106	Jeans	40
107	Python Programming Book	25
109	Mixer Grinder	12

8 rows selected.

7.2 Display Unavailable Products

```
SQL> SELECT
  2     p.Product_ID,
  3     p.Product_Name,
  4     i.Available_Stock
  5 FROM Product p
  6 JOIN Inventory i
  7 ON p.Product_ID = i.Product_ID
  8 WHERE i.Available_Stock = 0;
```

PRODUCT_ID	PRODUCT_NAME	AVAILABLE_STOCK
108	Data Science Book	0
110	Electric Kettle	0

8.Inventory Status Reports

8.1 Complete Inventory Report

SQL> SELECT
1 p.Product_ID,
2 p.Product_Name,
3 s.Seller_Name,
4 i.Available_Stock,
5 i.Warehouse_ID,
6 i.Last_Updated
7
8 FROM Product p
9 JOIN Seller s
10 ON p.Seller_ID = s.Seller_ID
11 JOIN Inventory i
12 ON p.Product_ID = i.Product_ID;

PRODUCT_ID	PRODUCT_NAME	SELLER NAME	AVAILABLE STOCK	WAREHOUSE_ID
101	Laptop	Tech World	10	201
102	Smartphone	Tech World	20	202
103	Headphones	Tech World	20	201
104	Smart Watch	Tech World	15	202
105	T-Shirt	Fashion Hub	50	204
106	Jeans	Fashion Hub	40	205
107	Python Programming Book	Book House	25	206
108	Data Science Book	Book House	0	207
109	Blender Grinder	Home Goods	12	208
110	Electric Kettle	Home Goods	0	209

10 rows selected.

8.2 Available and Unavailable Product Count.

```
SQL> SELECT 'AVAILABLE' AS STOCK_STATUS, COUNT(*) AS PRODUCT_COUNT FROM Inventory WHERE Available_Stock > 0 UNION ALL SELECT 'UNAVAILABLE' AS STOCK_STATUS, COUNT(*) AS PRODUCT_COUNT FROM Inventory WHERE Available_Stock = 0;
```

STOCK_STATU	PRODUCT_COUNT
AVAILABLE	8
UNAVAILABLE	2

8.3 Total Stock by Seller

```
SQL> SELECT
  2     s.Seller_ID,
  3     s.Seller_Name,
  4     SUM(i.Available_Stock) AS Total_Stock
  5 FROM Seller s
  6 JOIN Product p
  7 ON s.Seller_ID = p.Seller_ID
  8 JOIN Inventory i
  9 ON p.Product_ID = i.Product_ID
 10 GROUP BY s.Seller_ID, s.Seller_Name;
```

SELLER_ID	SELLER_NAME	TOTAL_STOCK
201	Tech World	75
202	Fashion Hub	90
203	Book House	25
204	Home Needs	12

8.4 Products with Zero Stock

```
SQL> SELECT
  2     p.Product_ID,
  3     p.Product_Name,
  4     i.Available_Stock
  5 FROM Product p
  6 JOIN Inventory i
  7 ON p.Product_ID = i.Product_ID
  8 WHERE i.Available_Stock = 0;
```

PRODUCT_ID
108
Data Science Book
0
110
Electric Kettle
0

PRODUCT_ID
PRODUCT_NAME
AVAILABLE_STOCK